

## **A2 – Reading Vocabulary – 6015, Jon Pulsipher**

**Hz** – A unit describing cycles per second, in Computer Science often used to describe how many instructions a processor can execute per second.

**Kilo-, Mega-, Giga-, Tera- (byte, Hz, etc)** – One thousand, one million, one billion, one quadrillion.  
 $10^3 * (1, 2, 3, 4...)$

**elf (in terms of Unix programs)** – Executable Linkable Format – a system used when assembling assembly code.

**Thrashing** – a state where RAM has been allocated in a problematic way where it needs to constantly open and page in info in a way that slows or stops processing.

**Virtual** – a helpful illusion in software where an interface or model is provided that appears to be access to a physical resource, but the details of how that physical resource are hidden by the provider of the virtualization. For example, the OS will provide a virtual picture of where addresses are assigned to a program that may not align with where that memory is physically.

**...Memory** – a place on hardware where a 0 or 1 is stored. When virtualizing, the place given to the program or user may not be the same as the address for the place in physical memory.

**...Address Space** – The memory space used when running a process with the Stack growing down from highest indices, the code at the lowest index, the BSS above that, and the Heap in some arbitrary place in the middle. This is a virtual place assigned by the OS corresponding to some space in real memory.

**...Address** – The number that corresponds to a virtual or physical location in memory.

**...Machine** – a virtual machine within a running computer is a simulated computer that (when running correctly) operates like an independent, physical computer would.

**Time Sharing** – To virtualize the processor so that multiple programs can seem to run at once, the OS will “Time Share”, switching between programs to give them dedicated processor time.

**Digital (Discrete) vs Analog** – In input, discrete values correspond to a single number whereas an analog input has a level of imprecision. A concrete example is a digital clock that displays numbers compared to an analogue clock where the hands have a number they’re closest to but never exactly on.

**Kernel vs OS** – The OS is the main manager of a computer and, as part of what it does, it talks to the Kernel which governs interaction with the hardware.

**Von Neumann** – The Von Neumann model of computing – a processor fetches instructions, decodes them, and executes them.

**Integrated Circuit (IC)** – Processors are made of these, composed of the tiniest circuits we can make that perform operations Electrical Engineers might be able to explain.

**State Machine (FSA)** – A “Finite State Machine” is a computation model that conceptualizes a computer as a machine that can be in one of a finite number of states at any point in time and will move from state to state based on inputs.

**Hardware (or program) privilege level** – We principally concern ourselves with the user privilege level (3) and the kernel privilege level (0).

**Context Switch** – When the processor switches to another process, it will save needed data so that it can store it when switching back, and load any data that had been saved for the context it is switching to.

**Lazy loading (of program, of memory, etc)** – a technique where only the part of the program that will be used is loaded. Used in web pages, on mobile devices to save power, and elsewhere.

**PID - Process ID** – a distinct number assigned to every process.

**DMA - direct memory access** – a feature that allows certain hardware subsystems to access memory without going through the CPU.

**MMU / TLB** – Memory Management Unit. It translates virtual addresses into physical addresses.

**Daemon** – a process that runs in the background, waiting to react to something. The sshd (ssh daemon) waits for ssh connections.

**Symbol (with respect to Code, object files, assembly)** – symbols are metadata that contain information about functions and global data, often a name, address, and filesize. Linkers use these to put together together a program.

**Shared Library** – libraries of code that multiple processes may use, but for efficiency they will be loaded into one memory space that associated processes have access to.